

2016.0 RANGE ROVER (LG), 501-27

FRONT END SHEET METAL REPAIRS

FRONT SIDE MEMBER AND SUSPENSION TOP MOUNT ASSEMBLY (G1454602)

REMOVAL AND INSTALLATION

REMOVAL

NOTE:

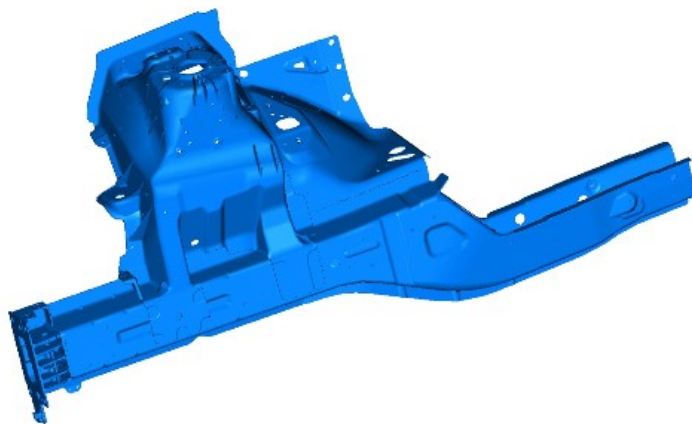
The front side member and suspension top mount assembly are installed in conjunction with:

- Front bumper cover
 - Front bumper
 - Front fender
 - Fender splash shield
 - Fender apron panel
 - Fender apron closing panel
 - Hood
 - Hood latch panel
 - Anti-lock brake system (ABS) module
 - Radiator
 - Condenser
 - Front subframe
 - Front suspension
 - Engine
 - Transmission
-

1.

NOTES:

- The assembly includes the deformation element, front side member to deformation element bracket, front side member, front side member closing panel, suspension top mount and the inner apron and wheelhouse panels.
- The right-hand front side member and suspension top mount assembly contains the stamped Vehicle Identification Number (VIN). When a new right hand front side member and suspension top mount assembly is requested it will be supplied stamped with the VIN. This operation is carried out within the Land Rover parts supply process.



E143269

The front side member and suspension top mount assembly is serviced as indicated.

-
2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).

-
3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.
- To make sure that the vehicle is correctly aligned, it must be placed on a Land Rover approved alignment jig.

-
4. For additional information, refer to: [Front Bumper Cover](#) (501-19 Bumpers, Removal and Installation).

-
5. For additional information, refer to: [Front Bumper](#) (501-19 Bumpers, Removal and Installation).

-
6. For additional information, refer to: [Fender](#) (501-02 Front End Body Panels, Removal and Installation).

-
7. For additional information, refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).

-
8. For additional information, refer to: [Fender Apron Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

-
9. For additional information, refer to: [Fender Apron Panel Closing Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and

Installation).

-
10. For additional information, refer to: [Hood](#) (501-02 Front End Body Panels, Removal and Installation).

-
11. For additional information, refer to: [Hood Latch Panel](#) (501-27 Front End Sheet Metal Repairs, Removal and Installation).

-
12. **NOTE:**

Right hand only.

For additional information, refer to: Anti-Lock Brake System (ABS) Module - RHD (206-09 Anti-Lock Control - Traction Control, Removal and Installation).

-
13. **NOTE:**

Left hand only.

For additional information, refer to: Anti-Lock Brake System (ABS) Module - LHD (206-09, Removal and Installation).

-
14. For additional information, refer to: [Radiator](#) (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal and Installation) / [Radiator](#) (303-03F Engine Cooling - TDV8 4.4L Diesel, Removal and Installation) / [Radiator - V8 N/A 5.0L Petrol](#) (303-03C Engine Cooling - V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol, Removal and Installation) / [Radiator - V8 N/A 5.0L Petrol](#) (303-03C Engine Cooling - V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol, Removal and Installation).

-
15. For additional information, refer to: [Air Conditioning System](#)

-
16. For additional information, refer to: [Front Subframe - TDV6 3.0L Diesel](#) (502-00 Uni-Body, Subframe and Mounting System, Removal and Installation) / [Front Subframe - TDV8 4.4L Diesel](#) (502-00 Uni-Body, Subframe and Mounting System, Removal and Installation) / [Front Subframe - V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol](#) (502-00 Uni-Body, Subframe and Mounting System, Removal and Installation).
-
17. For additional information, refer to: [Shock Absorber and Spring Assembly](#) (204-01 Front Suspension, Removal and Installation).
-
18. For additional information, refer to: [Engine](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Removal) / [Engine](#) (303-01E Engine - TDV8 4.4L Diesel, Removal) / [Engine](#) (303-01C Engine - V8 N/A 5.0L Petrol, Removal) / [Engine](#) (303-01D Engine - V8 S/C 5.0L Petrol, Removal).
-
19. For additional information, refer to: [Transmission - TDV6 3.0L Diesel](#) (307-01B Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Removal) / [Transmission - TDV8 4.4L Diesel](#) (307-01B Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Removal) / [Transmission - V8 N/A 5.0L Petrol/V8 S/C 5.0L Petrol](#) (307-01B Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed Automatic Transmission AWD, Removal).
-
20. For additional information, refer to: [Front Row Seat](#) (501-10B Seating - Long Wheel Base, Removal and Installation).
-
21. Remove the front floor carpet.
-

22.

NOTE:

Right hand only.

For additional information, refer to: [Brake Pedal and Bracket](#) (206-06 Hydraulic Brake Actuation, Removal and Installation).

23.

NOTE:

Right hand only.

Remove the front to rear brake pipes.

24.

NOTE:

Left hand only.

Remove the brake pipes at the bulkhead.

25.

NOTE:

Left hand only.

Remove the fuel supply and return pipes.

26.

NOTE:

Right hand only.

For additional information, refer to: [Fuel Fired Booster Heater](#) (412-

27.

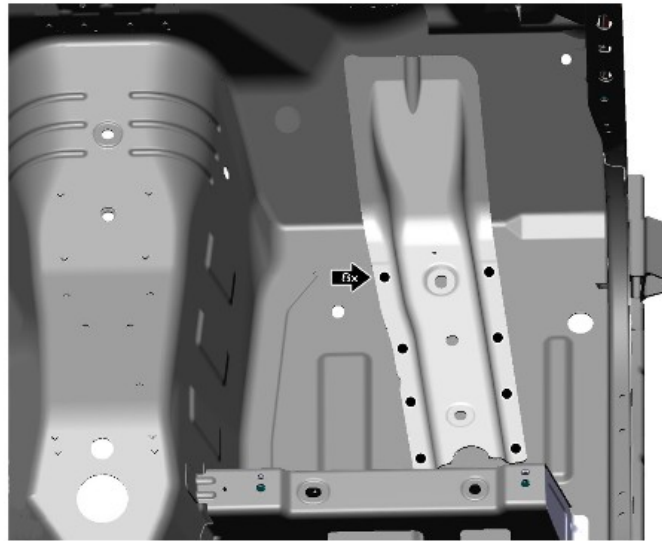
NOTE:

Right hand and left hand.

Release and position the front side member and suspension top mount wiring harness to one side.

28. Prior to removal, mark the position of the front side member and suspension top mount assembly in relation to adjacent panels for ease of alignment on installation.

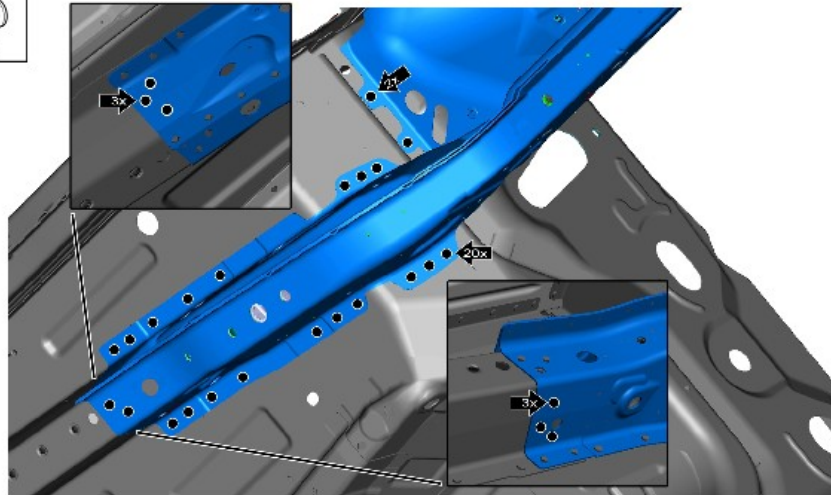
29.



E 143695

Remove the SPR's as indicated.

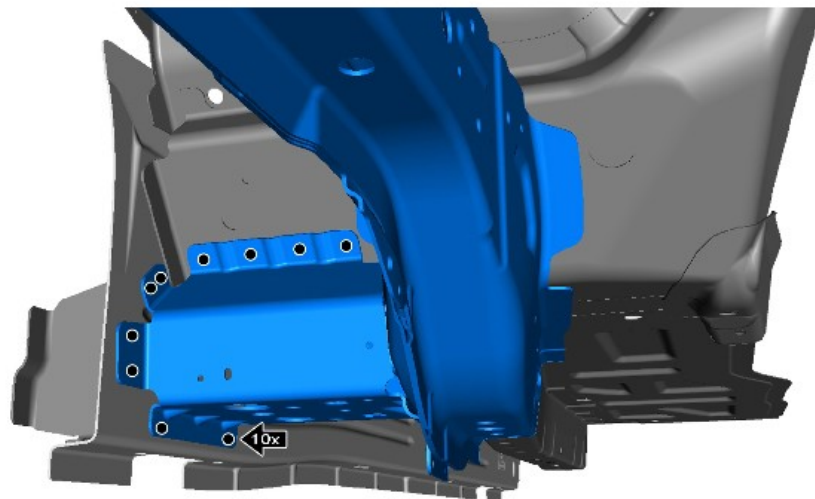
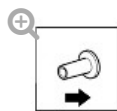
30.



E143276

Remove the SPR's as indicated.

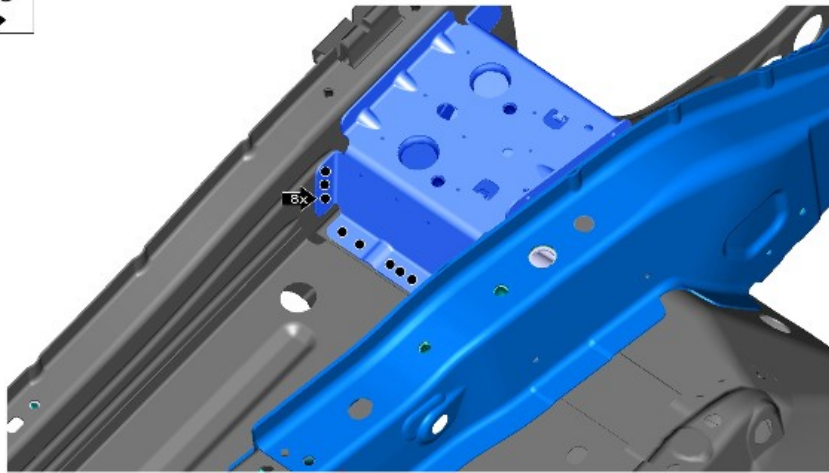
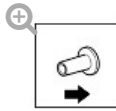
31.



E143273

Remove the SPR's as indicated.

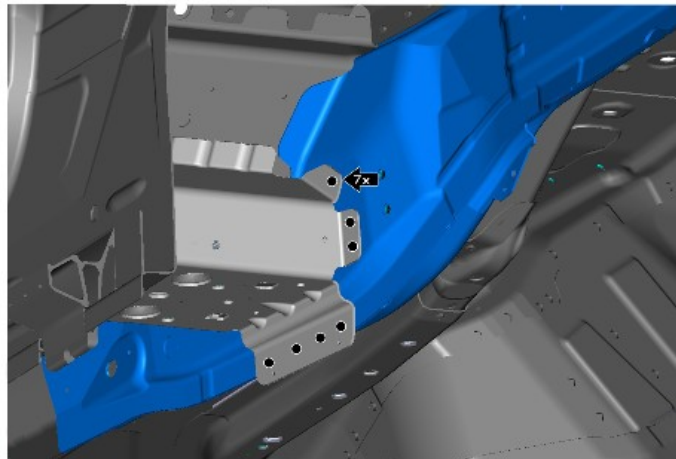
32.



E143691

Remove the SPR's as indicated.

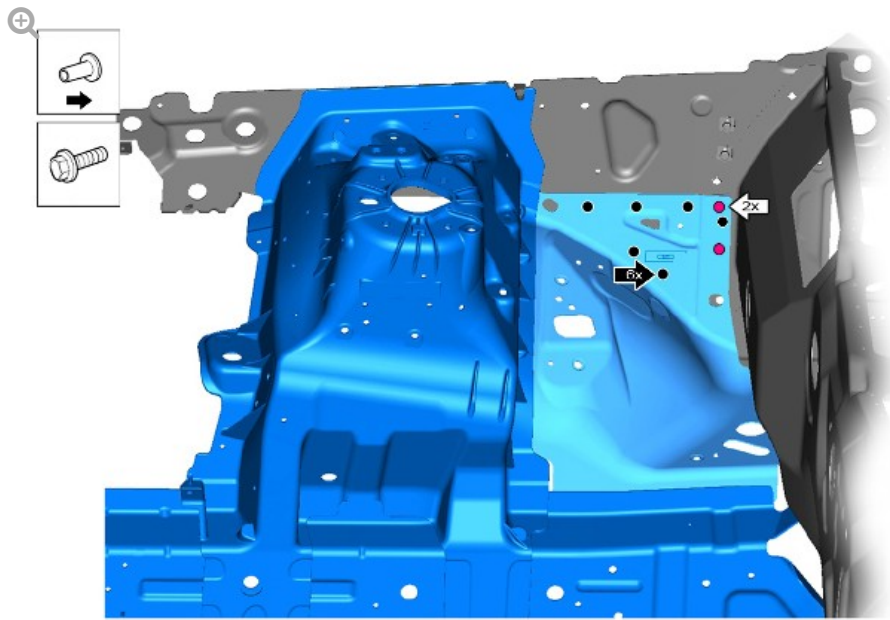
33.



E143274

Remove the SPR's and the reinforcement as indicated.

34.



E143271

Remove the SPR's and the retaining screws as indicated.

35.

CAUTION:

This step requires the aid of another technician as the front side member and suspension top mount assembly is heavy.

NOTE:

Retain the old panel as it will be required for reference on installation.

Separate the joints and remove the front side member and suspension top mount assembly.

INSTALLATION

CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

1. Remove rivet remnants.

2. Dress flanges where necessary.

3. Using a fine bristle disc clean and prepare the panel surfaces.

4.

NOTE:

Where there is limited or no access to drill through both the old and the new front side member and suspension top mount assembly when they are offered up to the vehicle assembly use the old front side member and suspension top mount assembly for reference. Measure and mark the locations for the installation of the BSF's on the new front side member and suspension top mount assembly.

Drill holes where BSF's are to be installed prior to offering up the replacement panel.

5. Deburr the drilled holes.

6.

CAUTION:

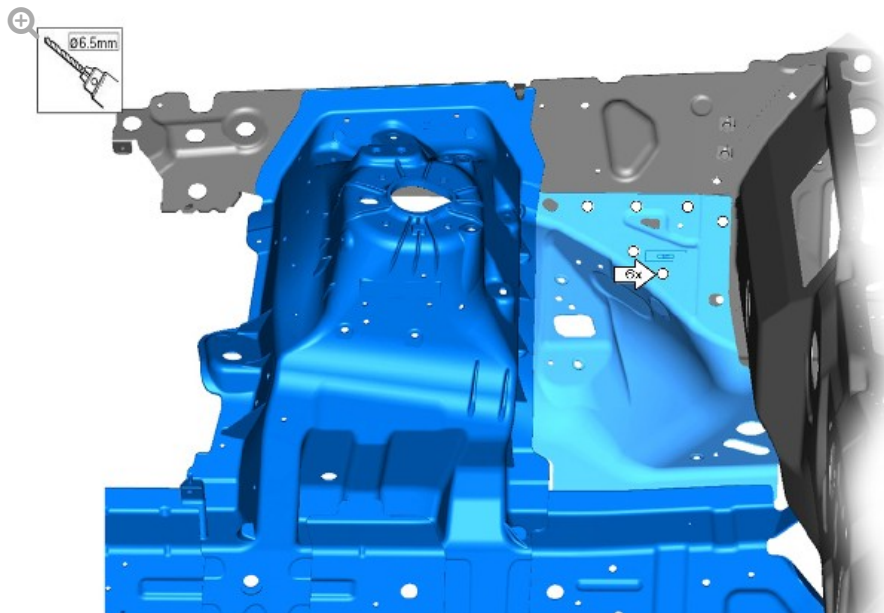
This step requires the aid of another technician as the front side member and suspension top mount assembly is heavy.

NOTE:

Make sure the alignment for the rivet nuts and drilled holes for the BSF's are correct.

Offer up the new front side member and suspension top mount assembly and clamp into position. Check alignment if correct proceed to next step, if not rectify and recheck before proceeding.

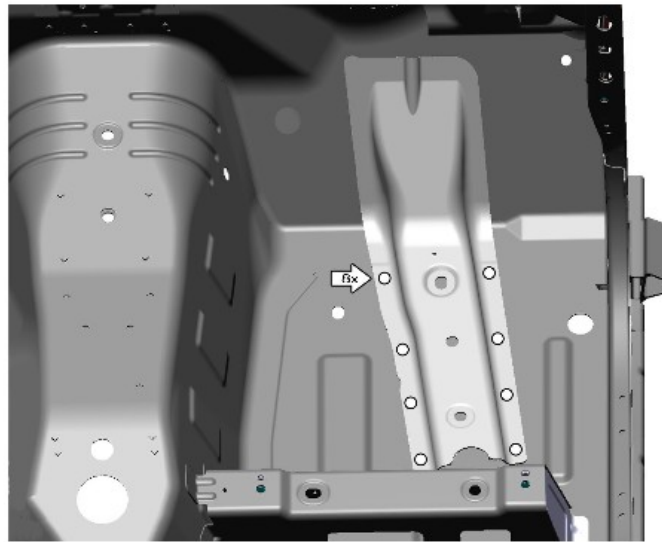
7.



E143270

Drill holes through the old panel into the new panel where BSF's are to be installed as indicated.

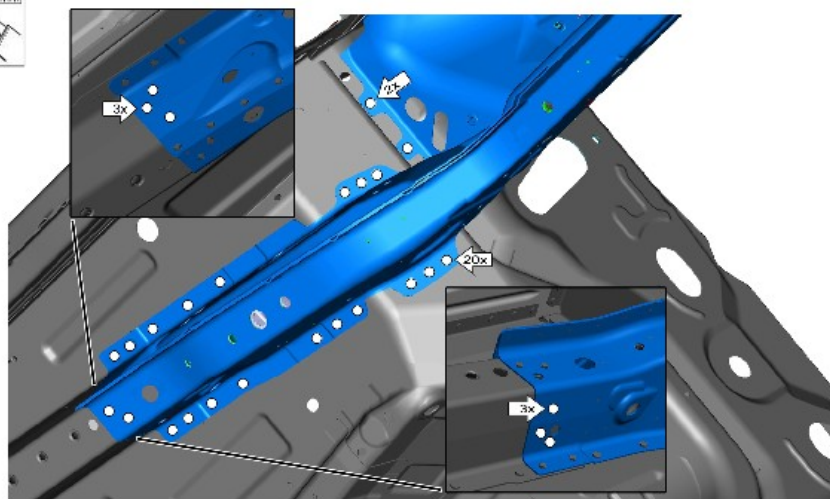
8.



E143697

Drill holes where the BSF's are to be installed as indicated.

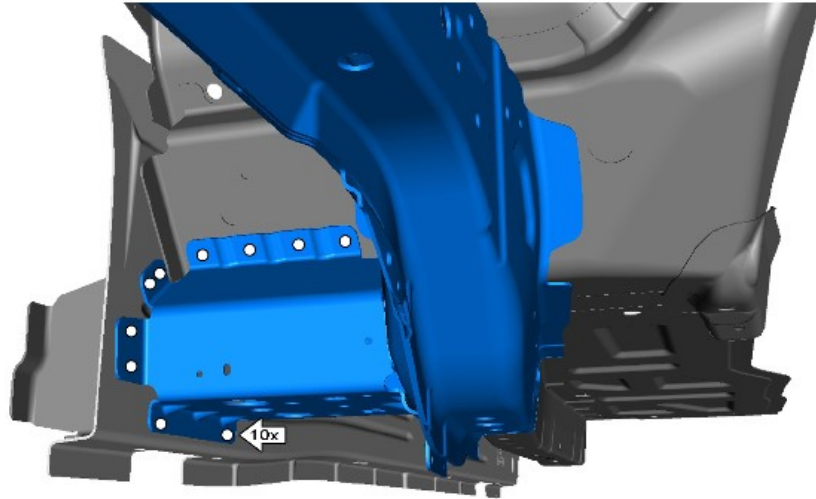
9.



E143277

Drill holes through the old panel into the new panel where the BSF's are to be installed as indicated.

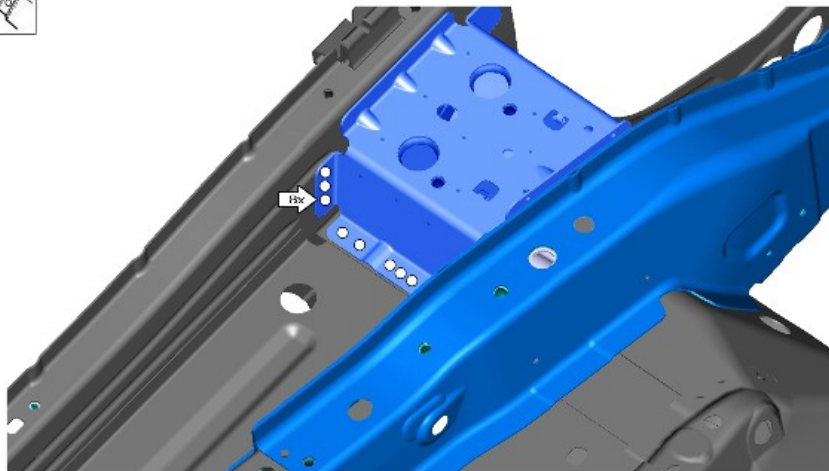
10.



E 143693

Drill holes through the old panel into the new panel where the BSF's are to be installed as indicated.

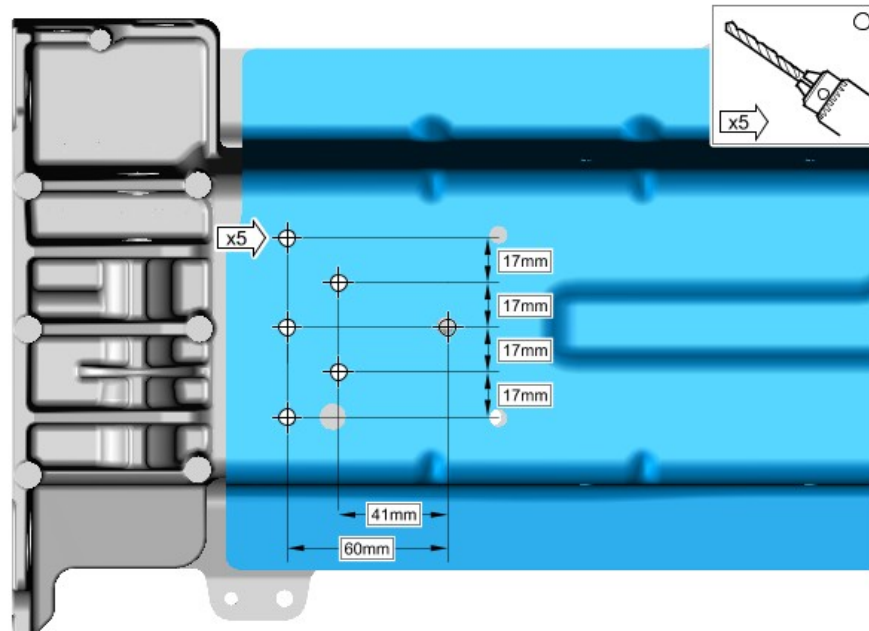
11.



E 143694

Drill holes through the old panel into the new panel where the BSF's are to be installed as indicated.

12.



E178384

Mark, measure and drill holes where the BSF's are to be installed as indicated.

13. Remove the new panel.

14. Deburr the drilled holes.

15. Using a fine bristle disc clean and prepare the panel surfaces.

16. Apply the coupling agent where the Land Rover approved bonding material is to be applied and allow to dry.

17.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.

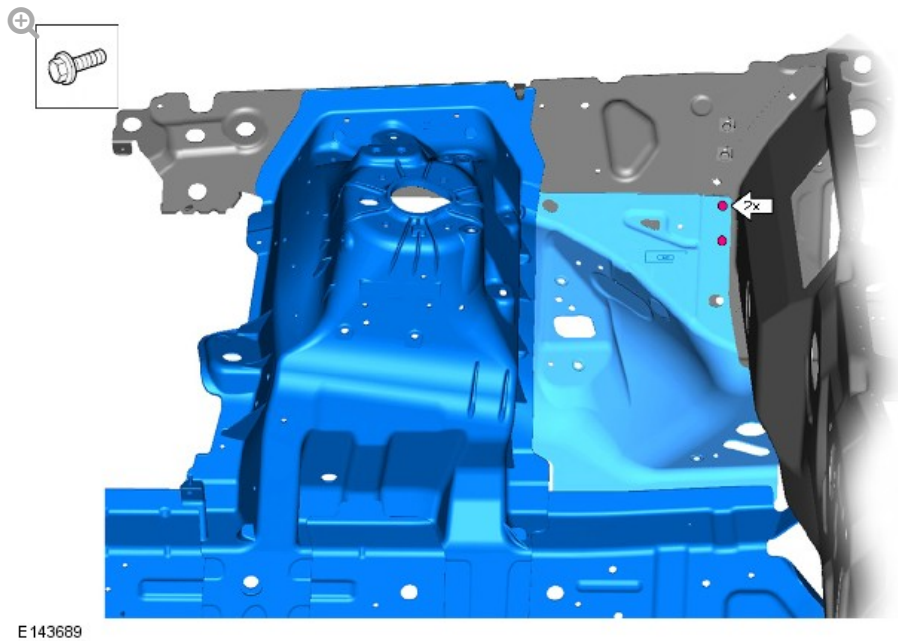
Apply a 5mm zig zag bead of Land Rover approved bonding material to the body joints in the area of the contact surface of the front side member and suspension top mount assembly.

18. Offer up the new front side member and suspension top mount assembly clamp into position. Check alignment if correct proceed to next step, if not rectify and recheck before proceeding.

19.

NOTE:

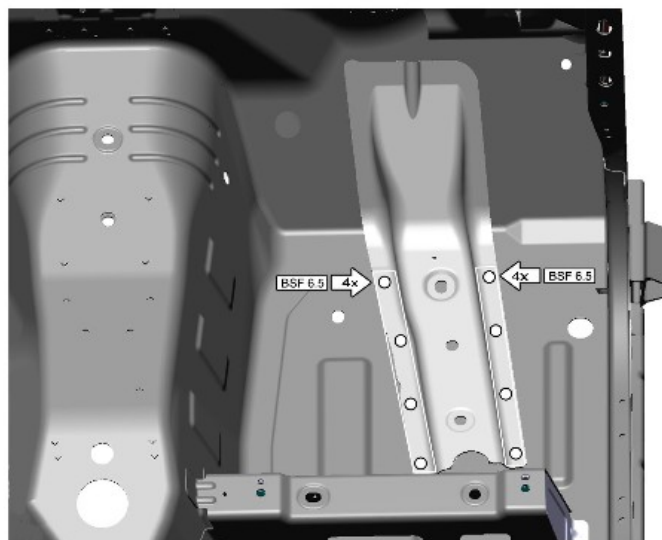
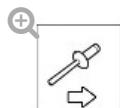
Do not fully tighten at this stage.



E143689

Install the retaining screws as indicated.

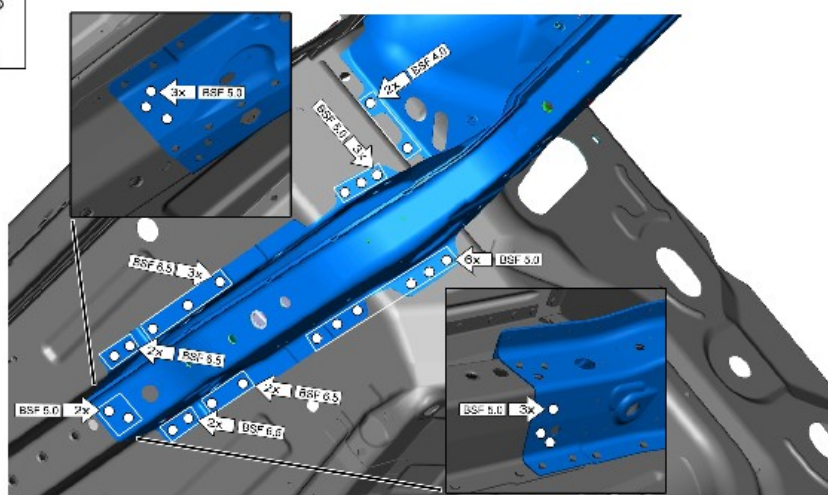
20.



E143696

Install the BSF's as indicated.

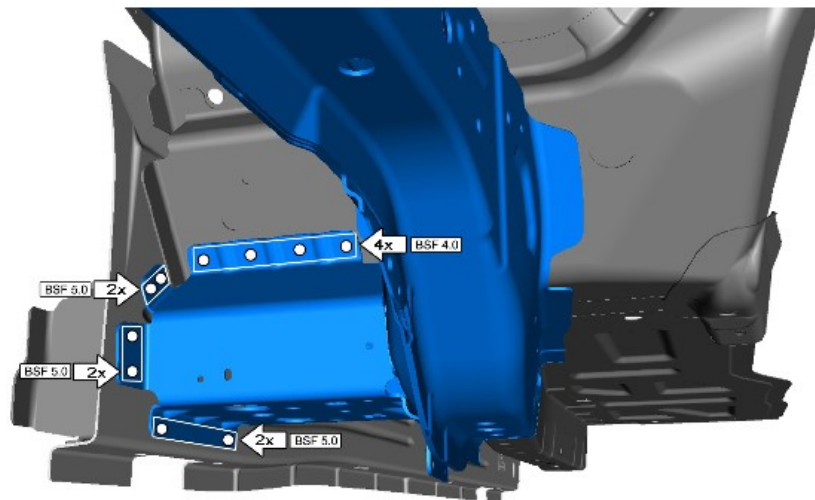
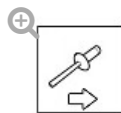
21.



E143690

Install the BSF's as indicated.

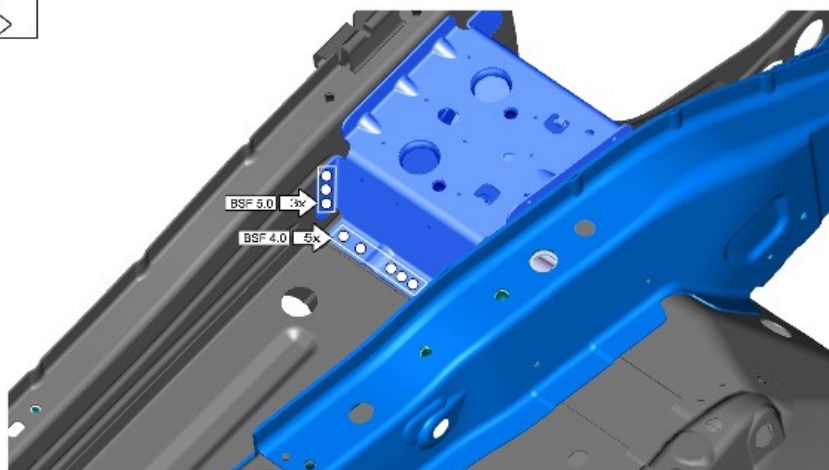
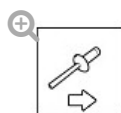
22.



E143278

Install the BSF's as indicated.

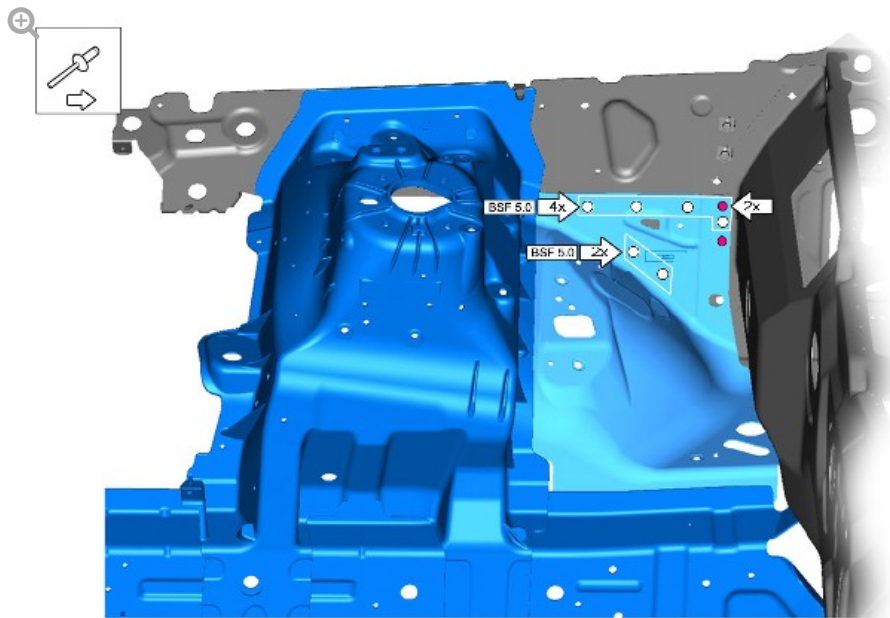
23.



E143275

Install the BSF's as indicated.

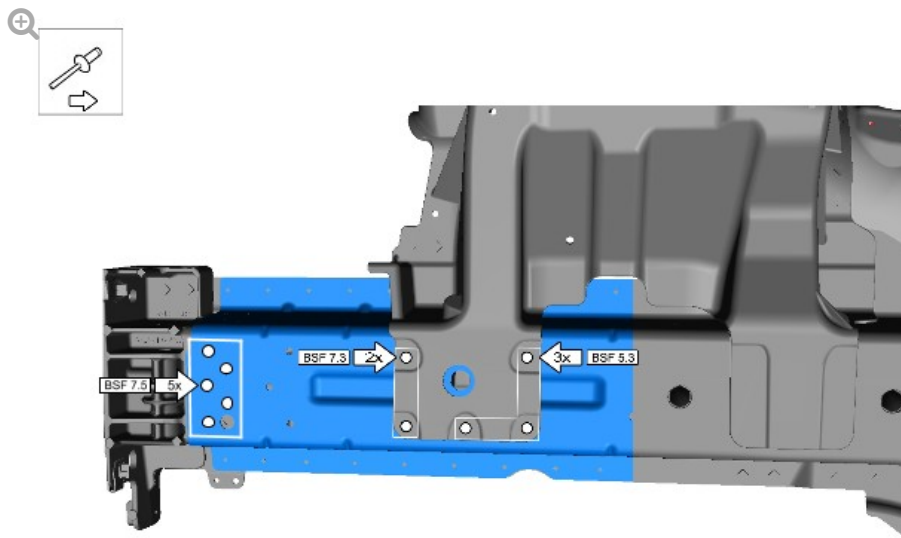
24.



E143688

Install the BSF's and tighten the retaining screws to 24Nm.

25.



E143687

Install the BSF's as indicated.

26. Remove any excess adhesive.

27. Make sure that any open or exposed panel joints are suitably sealed.

28. Make sure corrosion protection is applied to all areas affected by repair.

29. The installation of associated panels and components is the reversal of removal procedure.